

# **Stepper Lithography System Preventative Maintenance Contract Statement of Work**

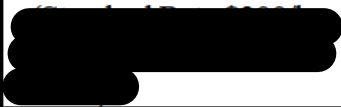
**PR:** \_\_\_\_\_

## **Background**

The GSFC Detector Development Laboratory (DDL) has a requirement for a maintenance contract for a Canon 5x i-Line Stepper Photolithography system (Stepper). The system is used to define submicron scale geometry structures in polymer based photoresists. The stepper aligns and exposes patterns through a reticle transferring it onto the wafer under fabrication. The exposure is completed using a step and repeat pattern where a subset of the wafer is exposed followed by a new exposure at the next position of the wafer. This patterning process is continued to cover the entire wafer. The stepper is suitable for patterning wafers of 100, 150 and 200 mm diameter and has high resolution accuracy, repeatability of feature geometry, and nanoscale registration accuracy. The stepper is an essential tool for the production of nearly all device fabrication projects in the Integrated Detector and Electronics Systems Branch Detector Development Laboratory at GSFC including the technology development and production of detectors for Cosmic Microwave Background Polarization technology, Microwave Kinetic Inductance Detectors, Far-Infrared Detectors, X-Ray Calorimeters, Large Format Magnetic Calorimeters, and Microfluidic Components, EUV and Far-IR/Sub-mm optical components, superconducting amplifiers, and MicroElectroMechanical (MEMS) devices which are important to Goddard's strategic focus areas including Planetary Defense, National Security and Moon to Mars Missions.

## **Requirements and Deliverables**

The requirements and deliverables of the preventive maintenance agreement is shown in the table below:

| Features                                                                                                                                                                                                    | Standard Conditions                                                                                                                  | Included                                                                                               |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------|
| Coverage Hours                                                                                                                                                                                              | Monday – Friday 9am-5pm<br>Excluding Canon Holidays                                                                                  | Yes                                                                                                    |
| FSE Service Support                                                                                                                                                                                         | Standard labor hours for preventative maintenance<br>(4) Quarterly PM visits per year.                                               | Yes                                                                                                    |
| Application Support                                                                                                                                                                                         | Standard labor hours for process, materials, apparatus, procedural, metrology, software or methods of optimization are not included. | Discounted Hourly Rate                                                                                 |
| Call Response                                                                                                                                                                                               | Guaranteed 4 hours<br>Target 2 hours                                                                                                 | Yes                                                                                                    |
| On site Response                                                                                                                                                                                            | Guaranteed 72 hours<br>Target 48 hours                                                                                               | Yes                                                                                                    |
| Additional labor hours within contract term with additional Purchase Order issued.<br>                                   | M-F days                                                                                                                             |                   |
|                                                                                                                                                                                                             | M-F after hours (when available)                                                                                                     |                   |
|                                                                                                                                                                                                             | Weekends - day shift (when available)                                                                                                |                   |
| Application and Software support available within contract term when requested and additional purchase order issued.<br> | M-F days “Standard” (when available)                                                                                                 |                   |
|                                                                                                                                                                                                             | M-F after hours “Standard OT” (when available)                                                                                       |                   |
|                                                                                                                                                                                                             | Weekends-day shift “Premium OT” (when available)                                                                                     |                   |
| Minimum Charge (additional FSE and/or Application Support)                                                                                                                                                  | 8 hours                                                                                                                              | 4 hours                                                                                                |
| Travel Labor Charge                                                                                                                                                                                         | 1/2 standard labor hour rate                                                                                                         | No charge for scheduled PM’s.<br>Maximum of 4 hours per call dispatch for unscheduled service repairs. |

|                                                              |                                                                                                                          |                                                                                                             |
|--------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------|
| Travel Expense<br>(hotel, car, lodging, etc.)<br>if required | At Cost                                                                                                                  | No charge for scheduled PM's.<br>For on call service repairs, travel<br>expense will be billed as incurred. |
| Urgent worldwide part<br>expedite fee [REDACTED]             | [REDACTED]                                                                                                               | [REDACTED]                                                                                                  |
| Consumable Parts                                             | Mercury and halogen lamps.<br>Y75-9174-022 or equivalent<br>2.7 KW HG lamp<br>GH0-5588-000 or equivalent<br>Halogen lamp | Included                                                                                                    |
| Repair parts                                                 | Any repair parts needed outside<br>of warranty coverage, require<br>additional purchase order.                           | [REDACTED]                                                                                                  |
| Spare Parts Usage Review                                     | Quarterly                                                                                                                | Yes                                                                                                         |
| Same / Next Day Ship Out                                     | Order must be received by noon<br>for same day ship out                                                                  | Yes                                                                                                         |
| Parts Returns<br>(no returns if opened by<br>customer)       | < 30 days                                                                                                                | Yes                                                                                                         |
| Parts warranty (# of days)                                   | Replaced by customer                                                                                                     | 30                                                                                                          |
| Parts warranty (# of days)                                   | Replaced by Canon                                                                                                        | 90                                                                                                          |
| Special Warranty Term (pro-<br>rated)                        | Up to 360 days warranty for<br>large units and assemblies                                                                | Yes                                                                                                         |
| Fee discount for Special Tools<br>Usage                      | Customer is responsible for<br>shipping charge                                                                           | 25%                                                                                                         |
| Service Update Notices                                       | General                                                                                                                  | Yes                                                                                                         |
| Service Update Notices                                       | Urgent                                                                                                                   | Yes                                                                                                         |

Canon will make commercially reasonable efforts to respond within the time frames set forth in the Service Plan.

1) Eligible Equipment

Equipment eligible for coverage under this contract shall be defined as: FPA-3030i5a Serial No. 1063105

2) Scheduled PM's

Canon agrees to provide Services outlines in this SOW to be mutually scheduled by the assigned project management from Canon and NASA for each event. The features listed in section 2) Service Plan are included in the cost of this SOW.

**Government-Furnished Equipment and Government-Furnished Information**

- none

**Place of Performance**

- NASA Goddard Space Flight Center Greenbelt MD USA.

**Period of Performance**

Period of performance is September 15, 2026, to September 14, 2027.